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Analysis of ALU Performance with Dynamic Voltage and Frequency Scaling Technique and Clock Gating

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Abstract. The Arithmetic Logic Unit (ALU) is a critical component of modern mad processors that handles arithmetic and logic operations. As power efficiency becomes more important in both high-performance and energy-constrained computing systems, Dynamic Voltage and Frequency Scaling (DVFS) and Clock Gating was emerged as approach for lowering power consumption. Therefore, this paper presenting the ALU design in both techniques, while maintaining acceptable performance. This paper shows the design finding of ALU performance with DVFS technique and Clock Gating technique. All this design was done using Synopsys simulation The study investigates how voltage and frequency variations affect the ALU's operational efficiency, power consumption, and overall processing speed. This study compares the effectiveness of three power-saving techniques, Dynamic Voltage and Frequency Scaling (DVFS), Clock Gating and Hybrid DVFS-Clock Gating Technique against a baseline with no power-saving techniques applied, in reducing design power consumption. The design power comparison shows that while DVFS increases the power consumption slightly to 53.66 μW, Clock Gating significantly reduces it to 32.28 μW compared to the baseline consumption of 52.30 μW. Meanwhile hybrid technique DVFS and Clock Gating shows good improvement with 54% of power reduction and time saving is 21% The power reduction comparison reveals that DVFS provides only a marginal 2.59%. However, the hybrid approach increases the complexity of design verification and testing. Ensuring the correctness of power state transitions and their impact on performance is more challenging than using a single technique.

1. Introduction

In VLSI, power management approach refers to a collection of techniques used to optimise power usage in integrated circuits and electronic systems. Power management is crucial in VLSI because of the growing need for energy-efficient products [1]. In VLSI, there are six main power management strategies supported. Initially, understanding power dissipation in VLSI is crucial. Power dissipation in VLSI circuits manifests in three primary forms: dynamic power, static power, and leakage power. Dynamic power arises from capacitive load. charging and discharging during

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switching operations. Static power results from leakage current flowing through transistors even when inactive. Leakage power emerges from sub-threshold leakage current and gate-oxide leakage [2]. Regarding power management techniques, various strategies exist, such as power gating, clock gating, voltage scaling, and dynamic voltage and frequency scaling (DVFS). Power gating involves selectively cutting off power to unnecessary or idle circuit blocks using switches or transistors, thus reducing static power consumption. Clock Gating disables the clock signal to specific circuits and sections during periods of inactivity, reducing dynamic power usage. Voltage scaling adjusts the supply voltage based on workload demands to conserve power, albeit needing careful implementation to avoid performance degradation and reliability issues. Frequency scaling tunes the clock frequency to balance performance and power consumption [3-4]. The Arithmetic Logic Unit (ALU) serves as a fundamental component of a processor, handling arithmetic and logical operations crucial for the performance of the system. Given its frequent operation, even minor optimisations in the ALU can yield substantial power savings at the system level. The ALU, being highly active in a processor, significantly contributes to dynamic power consumption due to its frequent switching actions. Due to its pivotal role in computation, the ALU becomes a key target for power optimisation strategies. Improvements in the ALU directly impact factors like processor clock speed, voltage scaling, and overall energy efficiency, making it a primary focus for power management research [3-7]. This paper explores power savings and performance testing using the ALU, focusing on the DVFS and Clock Gating techniques. A novel approach combining DVFS and Clock Gating is proposed to leverage the strengths of both techniques, aiming for improved power-performance trade-offs in complex designs. Implementing a hybrid technique that integrates both DVFS and Clock Gating mechanisms optimises power savings while maintaining satisfactory performance levels.

2. Literature Review

2.1 ALU Performance and Efficiency

Arithmetic operations conducted within an ALU are fundamental mathematical computations carried out on binary numbers. The ALU assumes the significant role of executing a range of arithmetic operations, including addition, subtraction, multiplication, and division. Each of these operations serves a distinct purpose and is intricately connected within the ALU's functionalities.

In the ALU may also set various flags or status bits, providing insights into the outcome of the operation. The zero flag, a critical indicator, indicates when the result of an operation is zero. This flag is significant for subsequent instructions involving equality testing or conditional branching based on this condition. The carry flag, another vital indicator, is activated when an arithmetic operation results in a carry or borrow. Conversely, the overflow flag is raised when a signed arithmetic operation surpasses the representable range, indicating a potential overflow impacting accurate representation within the given bit range. The auxiliary carry flag is relevant in binary-coded decimal arithmetic operations, indicating a carry out or borrow out from lower to higher bits. It is commonly used in decimal arithmetic operations and conversions. The parity flag, determined by the number of set bits in the result, is set if the result contains an even number of set bits; otherwise, it is cleared. This flag is useful for error detection or parity checking. Integrating these flags allows for the implementation of conditional statements, control loops, and various handling mechanisms, facilitating efficient decision-making and control flow within the CPU, thus significantly contributing to the computer system's overall functionality [9-11].

In designing the ALU, both performance and timing are paramount factors affecting the unit's overall efficiency. Performance, synonymous with the speed and efficiency of ALU operations, depends on several key factors. Clock frequency plays a crucial role in determining the ALU's

operational speed. A higher clock frequency generally leads to faster operations, but it may also introduce challenges related to power consumption and timing. The ALU's data width is another performance determinant, as a wider data path enables parallel processing of larger data chunks, resulting in faster execution [12,13].

Propagation delay, representing the time for a signal to traverse the ALU circuitry, is vital for data processing speed and output generation. Minimising propagation delay is crucial for achieving high-performance operations. Set and hold times define the period during which input signals must remain stable for correct operation, and violating these timing constraints can result in errors. Adherence to specific timing constraints set by the overall system design, including setup and hold times, maximum operating frequency, and critical parameters, is essential for proper ALU operation [14-15]. Failure to comply with these constraints can lead to incorrect operation or unpredictable behaviour. Optimising performance and timing in an ALU require a meticulous approach to architectural design, circuit implementation, and timing analysis. Designers aim to balance performance requirements, timing constraints, power consumption, and area utilisation to achieve an efficient and reliable ALU design. Design variations encompass a range of architectural choices and configurations that impact the ALU's functionality, performance, power consumption, area utilisation, and cost. Multiplexers and decoders within the ALU can be tailored based on required functionality and input/output configurations to optimise performance, minimise area, or enhance power efficiency. Another critical design variation involves choosing between combinational or sequential logic in the ALU design [16-19]. Combinational logic generates output solely based on current input values, while sequential logic relies on both current and past inputs. The selection between these approaches is influenced by specific ALU requirements and desired trade-offs in performance, complexity, and power consumption.

2.2 DVFS Implementation

Power management techniques in VLSI design require a comprehensive understanding of the system requirements, power profiles, and trade-offs between power consumption, performance, and area utilization. Power analysis evaluates power usage at various design levels, identifying power intensive components and discerning power distribution [20].

DVFS involves dynamically adjusting the voltage and frequency of a processor or system based on workload demands [21]. This adjustment creates a direct relationship between voltage and frequency, where changes in one parameter often necessitate corresponding adjustments in the order. In a digital circuit, power consumption is chiefly influenced by two factors which are voltage and frequency values [22]. The power consumed is calculated by the formula in equation (1).

$$P = C \times V^2 \times f \tag{1}$$

where C represents capacitance, signifying the activity or switching activity of the circuit. Considering the power formula, power consumption is directly proportional to the square of the voltage. Therefore, reducing the power supply voltage results in a quadratic reduction in power consumption. Lowering the voltage significantly reduces dynamic power dissipation caused by charging and discharging capacitive loads [23]. However excessively reducing voltage can lead to unreliable circuit operation and increased susceptibility to noise.

Power consumption is also directly proportional to the frequency. Higher frequencies demand more power for computations and instruction execution within a given timeframe [24]. By reducing the clock frequency, circuit power consumption can be decreased. However, lowering the frequency excessively can result in reduced performance and increased execution time for

tasks. The voltage and frequency relationship in DVFS allows for power performance trade-offs. Prioritizing power efficiency involves reducing voltage and frequency to minimize power consumption, sacrificing performance for energy savings and improved battery life. Conversely, when higher performance is essential, increasing voltage and frequency enhances computational capabilities at the expense of increased power consumption [25].

The voltage and frequency relationship in DVFS allows for power optimization while meeting performance requirements. By reducing the voltage and frequency, the associated power dissipation and heat generation can be lowered. This proactive thermal management helps prevent overheating, ensuring the system operates within acceptable temperatures limits and promoting reliability and longevity by preventing thermal induced failures [26].

The system can dynamically adjust the voltage and frequency to adapt to varying workload demands, optimizing performance. During periods of low computational intensity or idle time, voltage and frequency can be reduced to conserve power without significantly sacrificing performance. Conversely, during increased workload, raising the voltage and frequency ensures timely execution and meets performance requirements. By adjusting the voltage and frequency of individual components, DVFS enables power optimization for each specific task or processing unit. This ensures efficient resource utilization and overall energy savings. The flexibility of DVFS allows for a balanced power performance trade off.

As transistor sizes shrink and power becomes a critical concern, DVFS offers a mechanism to optimize power consumption without compromising performance. This future proofing capability ensures that VLSI systems can adapt to new technologies and remain power efficient over time. The dynamic adjustment of voltage and frequency optimizes power consumption, enhances system reliability, and enables efficient operation across a wide range of applications [27,28].

3. Methodology

It starts with designing the conventional 16-bit ALU with 10 different operations using Synopsis tools. This 16-bit ALU is a based model to integrate with others technique to optimize the performance of the circuit. This project test ALU performance in Clock Gating technique, DVFS technique and hybrid for DVFS-Clock Gating technique. Clock Gating significantly reduces power consumption but slightly increases timing. DVFS offers a balanced improvement in both power and timing. Hybrid DVFS and Clock Gating combines the strengths of both techniques, providing optimal power savings with acceptable performance trade-offs.

3.1 Clock Gating

For Clock Gating technique, the design is implemented using Synopsis as in Figure 1. Clock Gating using an additional register for each input operand which are 'gated_operandA' and 'gated_operandB'. The 'enable' input signal is used to control the update of these gated operands. When 'enable' is high, the input operands are captured and stored in the gated registers on the positive edge of the clock. The ALU operation is then performed on the gated operands based on the 'opcode' input. The result is stored in the 'result' register on the positive edge of the clock. During the simulation, the ALU will perform the requested operations on the given operands when the 'enable' signal is active. The computed results are available in the 'result' output signal. The result of simulating the testbench for the clock gated ALU will depend on the specific conditions under which the Clock Gating is applied. The Clock Gating logic disables the clock signal to the ALU when it is not actively processing data, reducing power consumption by minimizing unnecessary switching activity. Figure 1 shows block diagram for ALU in clock gating.

Figure 1 shows block diagram for ALU in clock gating

3.2 DVFS Module

The DVFS is calculated based on the parameters obtained from the ALU design analysis as shown in Figure 2. The modified ALU design is compiled with its testbench and the functionality of the ALU is observed and analysed based on its functional waveform. Operating point selection in DVFS refers to the process of determining the optimal combination of voltage and frequency settings for a given workload, aiming to find the most efficient operating point that minimizes power consumption without affect performance requirements.

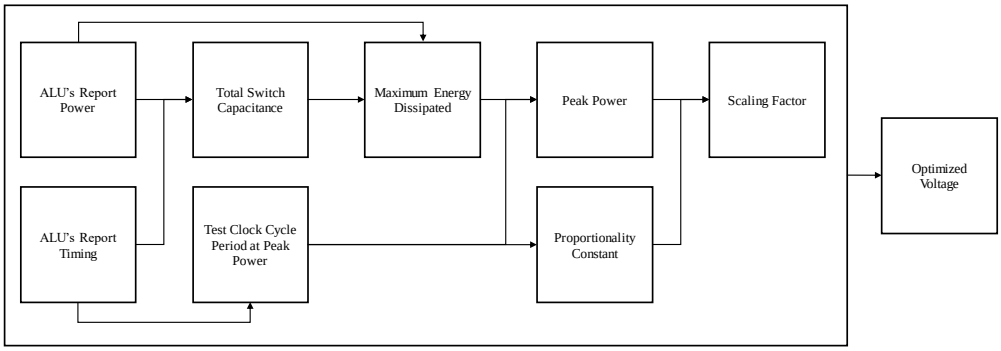

Figure 2: Block Diagram of DVFS

The DVFS system continuously monitors workload characteristics, such as computational intensity, memory access patterns, and task dependencies. The process involves a trade-off between power consumption and performance where lowering the voltage and frequency reduces power but result in slower execution and performance degradation. Maximum Energy Dissipated are based on capacitive load and voltage supply which can be calculated based on the equation (2). It shows reducing V_{DD} directly lowers energy consumption.

$$E_{MAX} = C_L \times {V_{DD}}^2 \tag{2}$$

The clock cycle period at peak power, T_{max} in the context of DVFS refers to the specific clock cycle duration during ALU design operates at its highest power consumption level as in equation (3).

$$T_{Max} = \frac{1}{f_{MAX}} \tag{3}$$

To expresses the relationship between voltage and timing which corresponds to DVFS needs. The proportionality constant can be calculated based on equation (4).

$$K = T_{Max} \times \frac{(V_{DD} - V_{TH})^{\alpha}}{V_{DD}} \tag{4}$$

The frequency scaling, f_s relation is given by equation (5).

$$f_s = k(V_{DD} - Vth)^{\propto} \tag{5}$$

Where k is a technology-dependent constant, Vth is threshold voltage α is an empirical constant (typically 1–2). From f_s, time scaling, Ts is determined. By dynamically adjusting V_{DD} and f, the power will be reduced while ensuring ALU operations complete within a required execution time.

Scaling factor is calculated based on comparing the minimum value of power and time. As for the power, maximum power is divided by dissipated power while for the time, the time scaling is divided by maximum time at peak power. To calculate the optimized voltage, the new proportionality constant needs to be identified by multiplying it with the new scaling factor value as in equation (6). From that, the related parameter can be inserted into the test time formula. Test time formula can be portrayed in two forms. Then, these two forms can be simultaneously calculated to obtain the test time and optimized voltage.

$$sf = \min\left[\frac{P_{max}}{P_{dissipation}}, \frac{Ts}{T_{max}}\right] \tag{6}$$

From the equation (6), optimize voltage, V_{DDopt} calculated from the original power supply, V_{DD} with scaling factor, *sf.* Through dynamic adjustment of voltage settings, DVFS allows systems to adapt to changing workload and requirements while maximizing energy efficiency. However, to calculate optimized voltage, the scaling time, *Ts,* need to be considered since optimized voltage is the unknown parameter which can be calculated through simultaneous equation.

3.3 DVFS with Clock Gating Design

A hybrid approach employed to integrate the benefits of DVFS with Clock Gating as in Figure 3. The hybrid strategy combines the processes of both strategies to maximize power savings while retaining acceptable performance. The ALU is designed as the core component where the hybrid power saving techniques integrate with Clock Gating is implemented on idle modules of the ALU to eliminate unnecessary clock activity. This step identifies specific in unactive modules in the ALU and disables their clock signals. For DVFS, the system dynamically adjusts voltage and frequency based on workload patterns. During periods of high activity, the system increases frequency for better performance, while during low activity, it reduces voltage and frequency to save power. Mathematical modelling for DVFS is reuse and equations are used to ensure the optimal balance between timing, power, and workload. Solving these equations fine-tunes the hybrid approach for better efficiency. The process concludes with the final optimized power and timing parameters for the ALU, ensuring that the hybrid technique delivers the desired results.

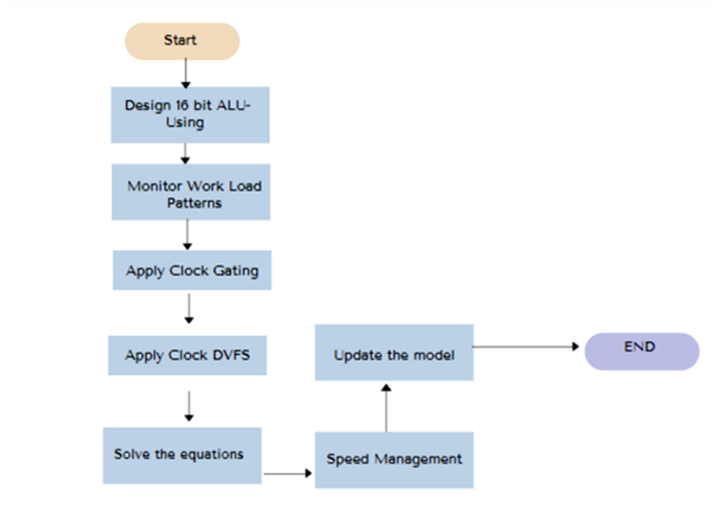

Figure 3: Flowchart for Hybrid DVFS-Clock Gating Technique

4. Result and Discussion

Simulating 6-bit ALU with 11 different operands involves running a testbench to provide specific inputs and observing the resulting outputs as in Figure 4.

Figure 4: 16-bit ALU Testbench Simulation Waveform

For the power consumed by conventional 16-bit ALU design. The global operating voltage is 0.7V while the voltage units is 1V. The cell internal power is 33.4528μW while the net switching power is 2.9000μW. The total dynamic power for the 16-bit ALU design is 36.3528μW while the cell leakage power is 15.9489μW. The total power consumed for a 16-bit ALU design is 52.3016μW.

The testbench for 16-bit ALU – Clock Gating integration design reported in Figure 5. For power, the global operating voltage is 0.7V while the voltage units are 1V. The cell internal power is 24.2431μW while the net switching power is 2.6876μW. The total dynamic power for the 16-bit ALU – Clock Gating design is 25.3535μW while the cell leakage power is 6.5462μW. The total power for 16-bit ALU – Clock Gating integration design is 32.2835μW.

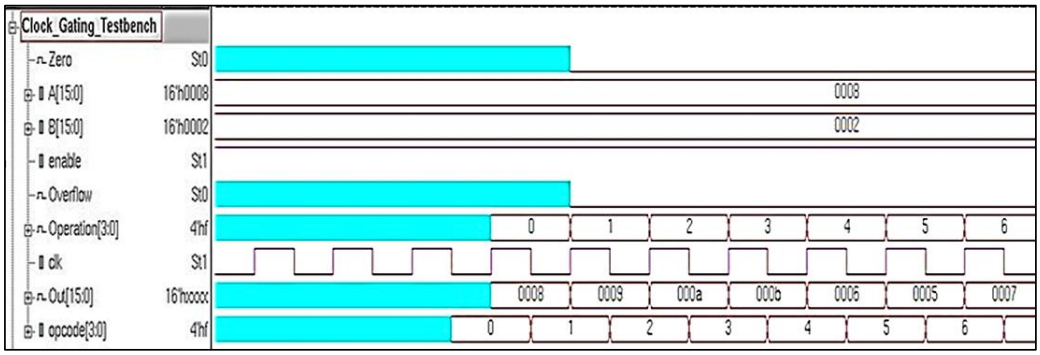

Figure 5: Clock Gating Testbench

By analysing the report power, the effectiveness of Clock Gating in reducing power consumption in the 16-bit ALU design can be assessed. Dynamic voltage and frequency scaling involves adjusting the supply voltage and clock frequency of the ALU in real time to optimize the trade-off between power consumption and performance. Since the library used in simulating the ALU is not supported to undergo DVFS simulation technique, the outcome is calculated based on its formula. The formula for calculating DVFS settings is often application-specific which suits the ALU and depends on several factors, including the characteristics of the circuit and the power – performance goals.

From the generated results from the simulated ALU design, several parameters can be found and use to calculate the total switched capacitance.

$Power = 52.3016\ \mu W$
$Voltage = 1.0V$
$Frequency = 50MHz$
$Time = 18.52\ ns$
$f_{MAX} = 53.9957\ MHz$
$Threshold\ Voltage = 0.8V$

The total power is divided by the total product of squared supply voltage and the frequency. Thus, it will produce the total switched capacitance which is 1.0460 pF. The maximum energy dissipated can be calculated by simply multiplying the total switched capacitance with the squared of supply voltage. The product of the equation will generate 1.0460 of energy. By following the time formula, the test clock cycle period at peak power can be easily calculated by dividing 1 with the maximum frequency. Hence, 18.52ns for T_{MAX} is obtained. From the obtained maximum energy dissipated and test clock cycle period at peak power, the peak power for the design can be calculated by dividing the E_{MAX} with the T_{POWER}. From that, 56.4795μW of peak power is obtained.

Proportionality constant is calculated by expressing the relationship between the test clock cycle period at peak power, supply voltage and threshold voltage. From the formula, $740.8\ \times 10^{-12}$ of proportionality constant for scaling factor equals to 1 is obtained. Scaling factor is calculated based on comparing the minimum value of power and time. As for the power, maximum power is divided by dissipated power while for the time, the time session is divided by maximum time at peak power. As for the power, 1.0798 is obtained while for time is 0.7436. Obeying the scaling factor formula which selects the minimum value generated, 0.7436 is selected to be the scaling factor value. To calculate the optimized voltage, the new proportionality constant needs to be identified by multiplying it with the new scaling factor value. From that, the related

parameter can be inserted into the test time formula. Test time formula can be portrayed in two forms. Then, these two forms can be simultaneously calculated to obtain the test time and optimized voltage, $V_{DDopt} = 0.9747V$ and $T_S = 17.5947\,ns$. The new optimize voltage and frequency from the test time is used. The value for the new power dissipated is 38.89 μW.

Table 1 shows the comparison power saving technique between none DVFS and DVFS technique. For conventional ALU with DVFS, power consumption without any power-saving technique is 52.30 μW. This serves as a reference for evaluating the effectiveness of other techniques. The design timing of 18.52 ns represents the time it takes for the design to complete a cycle.

Table 1. Comparison Power Saving Techniques

Power Saving Technique	None	DVFS	Clock Gating	Hybrid Technique
Design Power (μW)	52.30	51.66	32.28	23.88
Design Timing (ns)	18.52	17.59	19.63	14.52
Power Reduction (%)	-	1.22%	38.27%	54.34%
Timing Reduction (%)	-	0.55%	-0.06%	21.59%

For DVFS, it **r**educes power to 51.6578 μW, representing a small but meaningful decrease in power consumption. By dynamically adjusting voltage and frequency based on workload requirements, DVFS minimizes power usage without significantly affecting system performance. Reduces timing to 17.5947 ns, showing a slight improvement in performance. Adjusting voltage and frequency not only saves power but also optimizes performance to a certain extent by adapting to the workload's requirements.

For Clock Gating result, the power consumed by 16-bit ALU with global operating voltage is 0.7V while the voltage units are 1V. The cell internal power is 24.2431μW while the net switching power is 2.6876μW. The total dynamic power for the 16-bit ALU – Clock Gating design is 25.3535μW while the cell leakage power is 6.5462μW. The total power for 16-bit ALU – Clock Gating integration design is 32.2835μW. The data required time for the 16-bit ALU design is 19.64ns while the data arrival time for the 16-bit ALU – Clock Gating integration design is also 19.63ns. Achieves the most significant power reduction to 32.2835 μW, a decrease of 38.3% compared to the baseline. Clock Gating disables the clock signal to idle modules, effectively eliminating dynamic power usage for those parts of the design. Increases timing to 19.63 ns, a 5.99% delay compared to the baseline. Although effective in reducing power, the additional delay is caused by the need to re-enable clock signals for idle modules when they are required again, leading to overhead.

From the result, Clock Gating significantly reduces power consumption but slightly increases timing. DVFS offers a balanced improvement in both power and timing. Hybrid DVFS and Clock Gating combines the strengths of both techniques, providing optimal power savings with acceptable performance trade-offs. Choosing the appropriate technique depends on the application's power and performance requirements. For maximum power efficiency, Clock Gating is recommended. For a balanced solution, DVFS is a better choice. The hybrid approach is ideal for advanced systems requiring versatility and efficiency.

5. Conclusion

The implementation of Dynamic Voltage and Frequency Scaling (DVFS) and Clock Gating techniques in a 16-bit Arithmetic Logic Unit (ALU) demonstrates significant improvements in power efficiency while maintaining acceptable performance levels. From the comparative analysis, Clock Gating achieves the most substantial power savings but introduces a slight timing

penalty. On the other hand, DVFS provides a balanced approach, offering reductions in both power and timing. A hybrid approach combining DVFS and Clock Gating can maximize power efficiency while maintaining acceptable performance, making it an ideal choice for advanced systems requiring versatility and efficiency. Ultimately, selecting the appropriate power management technique depends on the specific application requirements. For maximum power efficiency, Clock Gating is recommended. For a balanced solution, DVFS is preferable. The hybrid approach is ideal for applications demanding both versatility and optimal power savings.

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